KOTLIN MULTIPLATFORM

# Building Nightly 

## The All-in-One Open-Source Music & Cinema Player




- Zero Ads & Tracking
- Android & Desktop
- In-App Cinema
- 28-Band FFT Audio
- AI Voice Search

A Complete Engineering Case Study & Showcase

github.com/aaweshdas/Audix-Multiplatform-Music-Player

THE PROBLEM

# Why Are Modern Media Apps So Frustrating?

### Invasive Ads & Clutter

Unskippable video and audio commercials interrupting playlists every 10 minutes.

### Paywalled Basic Features

Background play, lyrics, offline downloads, and seeking locked behind subscriptions.

### Fragmented Catalogs

Forced to juggle Spotify for music, YouTube for indie tracks, and piracy-riddled sites for cinema.

### Resource Bloat & Telemetry

Heavy Electron wrappers eating 2GB+ RAM with aggressive background tracking.

### The Solution:

Build an open-source, privacy-first client that does it all under one roof.

ARCHITECTURE

# One Codebase. Native Performance.

## ANDROID CLIENT

- AndroidX Media3 (ExoPlayer)
- Foreground Service Playback
- Lock Screen Notifications
- Hardware Media Key Hooks
- Dynamic Material 3 You Theming
- Low Battery & Memory Profile
- Min SDK 24 up to Android 15

## DESKTOP CLIENT

- VLCJ Native libVLC C-Bindings
- Lossless Gapless Audio Engine
- Windows, macOS & Linux Support
- Multi-Window Video Playback
- 28-Band Real-Time Visualizer
- Standalone Portable Zero-Install
- Hardware GPU Acceleration

### 90%+ Shared Kotlin Code

State machines, SQLDelight database, network pipelines, and UI layouts are completely shared across

CAPABILITIES

# Universal Music & In-App Cinema

### Decoupled Extension SPI

Plug-and-play architecture pulling tracks directly from Spotify, YouTube Music, JioSaavn, SoundCloud, and RadioBrowser with automatic fallback mirrors.

### In-App Cinema Theater

Embedded streaming studio with automated Cinemeta IMDb resolution (`tt...`), episode indexes, and multi-server stream mirrors.

### Sandboxed Security Shield

Custom WebView interceptor blocking unwanted ad redirects, popups, and intrusive third-party scripts.

### Synced Synchronized Lyrics

Real-time scrolling karaoke lyrics powered by LRCLIB, Genius, and Musixmatch.

DEEP TECH

# Audiophile Visualizer & AI Voice

## 28-Band Cooley-Tukey Radix-2 FFT

- Mathematical Fast Fourier Transform computed over raw 512-point audio chunks.
- Hann windowing applied to eliminate edge spectral leakage.
- Logarithmic frequency scaling across 28 bands (20 Hz sub-bass to 20 kHz treble).
- Hardware-accelerated rendering at fluid 60fps.

## Dual AI Speech-to-Text Engines

- Whisper Large V3 Turbo running via Groq for ultra-low-latency transcription.
- Deepgram Nova-2 fallback with real-time Voice Activity Detection (VAD).
- Hands-free voice querying for songs, artists, albums, and cinema movies.
- Zero audio storage or tracking — privacy by design.

GET STARTED

# 100% Free & Open-Source

Nightly is ready for you to explore, test, and contribute.

## Check Out The Repository

**https://github.com/aaweshdas/Audix-Multiplatform-Music-Player**

 Star the repo on GitHub if you love KMP & open source

 Download the Android APK or Windows Portable Zip

 Clone the project and run locally in Android Studio / IntelliJ

 Connect with me on LinkedIn to discuss KMP & mobile tech!

## Crafted with passion by Aawesh Das

Software Engineer & Cross-Platform Enthusiast

Tech Stack: Kotlin Multiplatform • Compose • ExoPlayer • libVLC • AI/STT

**Drop a comment or share this with anyone exploring Kotlin Multiplatform! **